

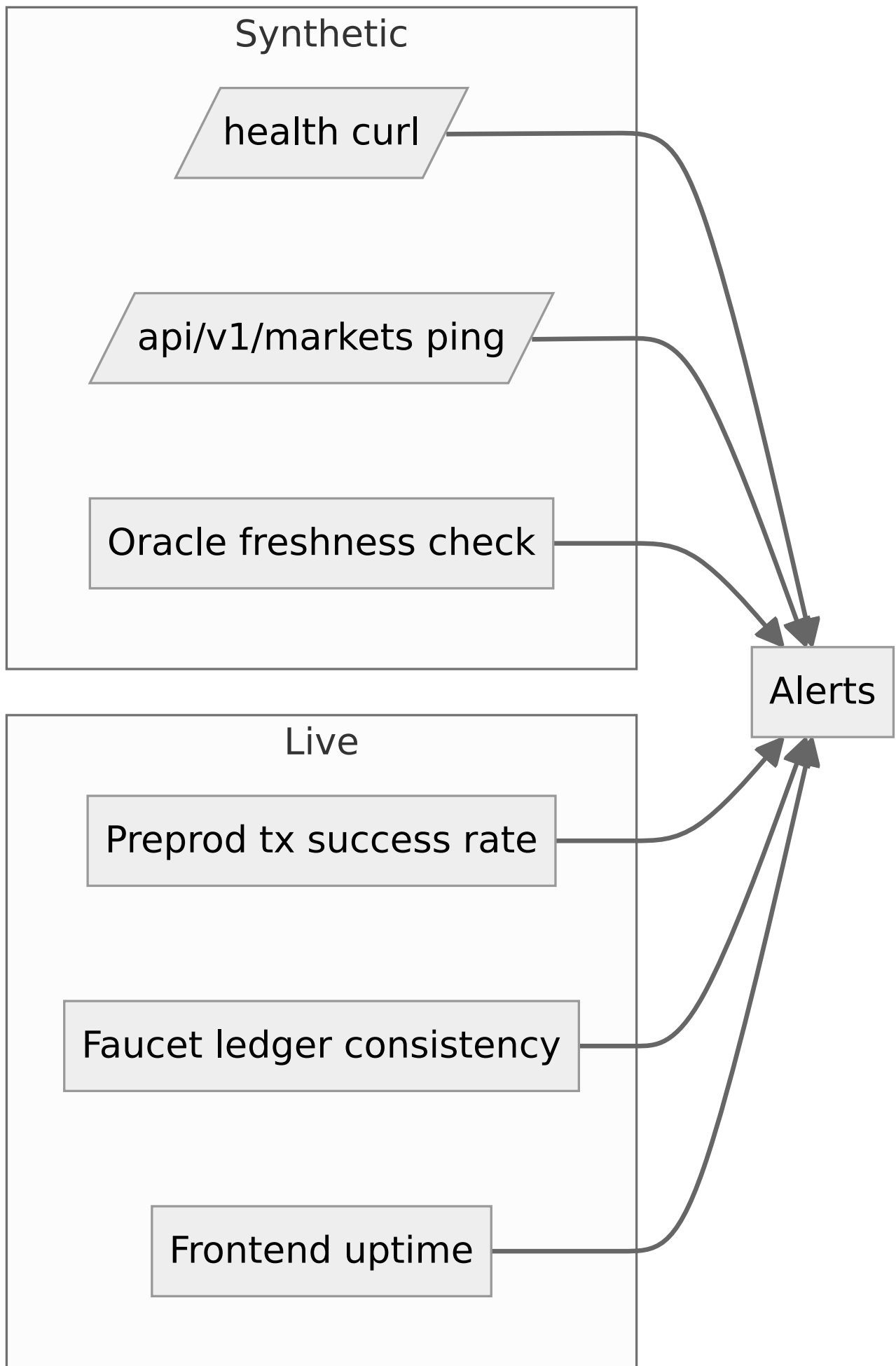
Pre-IPO Perps — Post-Launch Operations

Repository: pre-ipo-perps (dextrlabs-dev/Pre-IPO-Perpetuals-Cardano) **Document status:** Final, version-tagged for project close-off. **Scope:** Monitoring, maintenance, incident response, and governance for the deployed Pre-IPO Perpetuals stack post-launch.

This document picks up where [FINAL-TECHNICAL-DOCUMENTATION.md](#) ends. It assumes the Preprod public demo is running per §3 of that document.

1. Monitoring

1.1 Layers + signals



Signal	Source	Check	Alert threshold
Backend health	GET /health	HTTP 200 + {ok: true}	Two consecutive failures in 60 s
Markets endpoint	GET /api/v1/markets	Three markets returned, each with a recent last_oracle_time	Any market absent for 5 minutes
Oracle freshness	On-chain last_oracle_time vs wall clock	$\text{now} - \text{last_oracle_time} \leq 2 \times \text{min_interval_ms}$	Stale beyond 2x window
Preprod tx success	Oracle / mint / redeem / swap submissions in logs	Submitted tx ↔ Blockfrost confirmation	Single submit failure → warn; two within 10 minutes → page
Faucet ledger consistency	offchain-platform-backend/data/usdcx-faucet.json	After every mint / redeem the ledger delta equals $\text{qty} \times \text{twav_price} / \text{scale}$	Any drift > 0.1 USDCx
Frontend availability	Synthetic browser ping	HTTP 200 + presence of #trader-dashboard root	Two consecutive failures
Reference-candle proxy	GET /api/v1/candles?market=...	HTTP 200 + non-empty array	Three consecutive failures

1.2 Logging

- **Backend** writes structured JSON lines: tx submit, oracle worker tick, faucet mutation, session-store update. Operators are expected to pipe these into their log aggregator of choice.
- **Frontend** logs API call failures to the browser console with the same correlation id the backend uses (X-Request-Id), simplifying support diagnosis.
- **Validator (on-chain)** is observable only through the chain: use cli/preprod-three-markets-report.ts to dump current datum + reserves for any market on demand.

1.3 Dashboards (recommended)

Although the project does not ship a bundled dashboard, an operator should provision:

1. A **rolling counter** of Oracle / Mint / Redeem / Swap tx submissions per market per hour.
2. A **timeseries** of current_price vs twav_price (sourced from the Preprod CSV or the markets endpoint).

3. A **gauge** of the operator wallet's Preprod tADA balance (Blockfrost addresses endpoint).
 4. **Alerting** wired into a paging system (PagerDuty, Opsgenie, Slack webhook).
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2. Maintenance

2.1 Routine

Caden ce	Task	Reference
Daily	Confirm GET /health + three markets present	§1.1
Daily	Confirm oracle freshness for all three markets	§1.1
Weekl y	Snapshot offchain-platform-backend/data/usdcx-faucet.json and session-store directory to off-host backup	§2.3
Weekl y	Verify operator wallet tADA balance $\geq$ 100 tADA on Preprod (fund from faucet if low)	https://docs.cardano.org/cardano-testnets/tools/faucet
Month ly	Run aiken check against validators/preipo.ak; redeploy if checks change	FINAL-TECHNICAL-DOCUMENTATION.md §5
Month ly	Re-run cli/preprod-three-markets-report.ts; diff against the last committed CSV; investigate any new failure rows	preprod-run-2026-03-23T10-48-17-980Z.csv
Per releas e	Re-render PDF docs (deno run -A docs/render-pdf.ts); tag final-vN.M	§3

2.2 Dependency hygiene

- **Aiken** — pinned in aiken.toml. Bump only after a Preprod re-run shows identical CSV semantics.
- **Lucid Evolution** — pinned in cli/preipo-evo-core.ts imports. Validator-affecting bumps must run the full Preprod cycle.
- **Deno / Node** — deno.json and package.json are the source of truth. Use deno cache --reload main.ts + npm ci to validate clean installs before each release.

2.3 Backup / restore

Artifact	Backup approach	Restore path
USDCx faucet ledger	Weekly snapshot of offchain-platform-backend/data/usdcx-faucet.json	File replacement + backend restart
Last-spend session store	Weekly snapshot of offchain-platform-backend/data/session-store/	Either file replacement or chain-driven re-derivation via cli/preprod-three-markets-report.ts
Preprod evidence CSV	Committed to repo on each refresh	git checkout historical version
Operator seed	Out-of-band secret manager (1Password, AWS Secrets Manager, etc.); never committed	Restore environment variable PREIPO_WALLET_SEED
Plutus blueprint	Committed (plutus.json); regenerable via aiken build	aiken build

3. Incident response

3.1 Severity levels

Severity	Definition	Examples
SEV-1	Trader-facing operations broken across all markets	API down, validator address rotated unexpectedly, operator seed compromised
SEV-2	Single market degraded, or oracle freshness lost	Oracle worker stalled, one market's tx submissions failing
SEV-3	Quality issue with no immediate trader impact	Stale faucet snapshot, slow charts, doc bug

3.2 Playbooks

SEV-1: Backend down (HTTP unreachable)

1. Confirm the backend process state on the host.
2. Tail logs for the last submit / oracle / faucet line.
3. Restart with `deno task serve:preprod`. If restart fails, roll back to the previous tagged release.

4. Notify users via the public status channel; do not alter validator state until root cause is identified.

SEV-1: Operator seed compromise

1. Rotate the operator seed in the secret manager.
2. Generate a fresh deterministic wallet, fund it from the Preprod faucet.
3. Re-run `cli/preprod-three-markets-report.ts` to seed new locks for each market; old script UTxOs remain spendable only by the prior seed (Preprod scope only — design implication for any future mainnet operation is captured in §4.3).

SEV-2: Oracle stalled on one market

1. Tail `oracle-worker.ts` logs for the affected `marketId`.
2. Verify the reference candle endpoint returns data (`GET /api/v1/candles?market=...`).
3. Manually submit a single oracle tick: `deno run -A cli/preipo.ts oracle <marketId>` (the CLI emits the same redeemer the worker would).
4. Re-enable the worker.

SEV-2: Faucet ledger drift

1. Stop the backend.
2. Replay the most recent mint / redeem txs from the CSV / chain; recompute the ledger delta per address.
3. Restore the corrected JSON from the most recent backup if reconciliation cannot complete cleanly.
4. Restart the backend.

SEV-3: Stale Preprod evidence

1. Re-run `cli/preprod-three-markets-report.ts`.
2. Commit the new CSV under `docs/preprod-run-*.csv` and update the embedded `PREPROD_LEDGER_CSV` in `preipo-frontend/src/data/preprodLedger.ts` if Trade history must reflect the new run.

3.3 Communication

- **Status page / channel:** A public Discord, Telegram, or status page is recommended; mention the affected market and the redeemer path.
- **Postmortems:** All SEV-1 and SEV-2 incidents get a written postmortem committed to `docs/postmortems/` within 14 days.

4. Governance

4.1 Decision-making

Decision class	Required approvers	Forum
Validator change (any edit to <code>validators/preipo.ak</code>)	Smart-contract engineer + 1 reviewer with on-chain audit experience	PR + Preprod evidence re-run
Redeemer-affecting off-chain change	Backend engineer + reviewer with <code>openapi.json</code> familiarity	PR + Preprod evidence re-run
Frontend release	Frontend engineer + UX reviewer	PR
Documentation change	Any contributor	PR
Operator wallet rotation	Project lead	Out-of-band ticket
Funding-window scope change	Project lead + Catalyst proposer	Catalyst public channel

4.2 Contribution flow

1. Fork or branch `main`.
2. Implement + run `aiken` check and the relevant test surface.
3. For redeemer-affecting changes, re-run `cli/preprod-three-markets-report.ts` and attach the fresh CSV.
4. Open a PR with: scope summary, evidence link, screenshots if UI, and `openapi.json` diff if API.
5. Reviewer approves; maintainer merges + tags if it's a release-impacting change.

4.3 Forward-looking design implications

The MVP intentionally keeps **trader authority** scoped to a single key (`PerpDatum.trader`) and **oracle authority** scoped to a single keyset hash (`MarketState.publisher_keyset_hash`). The following expansions are out-of-scope for this funding round but should be planned for any subsequent phase:

- **Multi-trader markets** — extend `PerpDatum.trader` from `PubKeyHash` to a script credential, gated by a separate access-control validator.
- **Distributed oracle keyset** — replace the single `publisher_keyset_hash` with a multisig redeemer check (e.g. $\geq K$ of N signatures), keyed by an on-chain keyset registry.
- **Native USDCx asset** — promote the off-chain faucet ledger to a real minting policy + token-policy redeemer that mirrors mint / redeem in value, retiring the JSON faucet.
- **Governance bridge** — anchor `validators/preipo.ak` parameters (`max_variance_bps`, `min_interval_ms`, `pool_deviation_bps`) inside a governance-controlled reference datum, so parameter changes do not require validator redeloys.

4.4 Compliance + legal

- The MVP is a **research / demonstration** stack on Cardano Preprod. It carries no real-money settlement, no fiat rails, and no securities-law obligations as deployed.
 - Any future mainnet deployment that lists synthetic exposure to non-public companies must be reviewed by jurisdiction-appropriate counsel; this is documented for visibility and is **not** a part of the current funded scope.
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5. Document control

Item	Value
Version	final-v1.0
Companion documents	MVP-TECHNICAL-DOCUMENTATION.md , FINAL-TECHNICAL-DOCUMENTATION.md , CLOSE-OUT-REPORT.md
Render	deno run -A docs/render-pdf.ts (point mdPath at this file for the PDF export)